

Liquid–Liquid Extraction (LLE) in Parallel Microflows

Group 10

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INTRODUCTION & PROBLEM SETUP

Liquid–liquid extraction (LLE) in microchannels relies on two immiscible fluids flowing in parallel, allowing a solute to diffuse from one phase into the other across a stable fluid–fluid interface. Such microscale systems offer high interfacial area, short diffusion lengths, and predictable laminar flow fields, making them attractive for efficient separations and chemical processing. The hydrodynamics of the flow regime directly influence mass transfer, and therefore an accurate description of both the velocity field and the concentration field is essential for predicting extraction performance.

This project reconstructs the main results of Picardo et al. (2014) for:

- 1) **1D stratified flow** between two infinite parallel plates,
- 2) **2D stratified flow** in a rectangular microchannel, and
- 3) **Core–annular flow** in a circular microchannel.

For each geometry, implementing the governing equations for fully developed laminar flow and steady mass transfer using the same nondimensional framework as the original paper. The analysis includes:

- computation of the **velocity profiles** using analytical expressions.
- derivation of the **advection–diffusion equation** for solute concentration,
- evaluation of **extraction metrics**, including the Extraction Ratio and Extraction Efficiency, and
- verification of global **mass balance** through residual and cumulative flux plots.

To ensure consistency with Picardo et al., the simulations use the same dimensionless groups such as Péclet number, viscosity ratio, diffusion ratio, flow–rate ratio, holdup and follow the same boundary and interface conditions. Replicating the figures from the research paper enables a clear comparison between geometries and provides physical insight into how flow configuration, interface location, and transport time scales govern extraction performance in microchannels.

NUMERICAL METHOD: METHOD OF LINES (MOL)

The Method of Lines (MOL) is a numerical technique used to solve partial differential equations (PDEs) by converting them into a coupled system of ordinary differential equations

(ODEs). This is achieved by discretizing only the spatial derivatives while keeping the axial coordinate continuous. The method is particularly suitable for advection–diffusion systems in microchannel flows, where transverse diffusion is strong but axial variation is smooth.

Overview and Concept

Consider a general advection–diffusion PDE of the form

$$v(y) \frac{\partial c}{\partial x} = D \frac{\partial^2 c}{\partial y^2},$$

defined for $0 \leq y \leq 1$ and $0 \leq x \leq L$.

The MOL proceeds in three steps:

- 1) **Discretize the transverse coordinate.** Choose N grid points $y_1, y_2, \dots, y_N$ and approximate the spatial derivatives using finite differences:

$$\left. \frac{\partial^2 c}{\partial y^2} \right|_{y_i} \approx \frac{c_{i+1} - 2c_i + c_{i-1}}{\Delta y^2}.$$

- 2) **Convert the PDE into a system of ODEs.** Substituting the finite–difference expressions into the PDE yields, at each grid point:

$$v_i \frac{dc_i}{dx} = D \frac{c_{i+1} - 2c_i + c_{i-1}}{\Delta y^2}, \quad i = 2, \dots, N-1.$$

This gives N coupled ODEs of the form:

$$\frac{d\mathbf{C}}{dx} = \mathbf{A}\mathbf{C},$$

where $\mathbf{C} = (c_1, c_2, \dots, c_N)^T$ and $\mathbf{A}$ is the discrete diffusion operator modified by the velocity field.

- 3) **Integrate along the axial direction.** The resulting ODE system is solved using an ODE solver (e.g., Runge–Kutta), with the inlet concentration profile as the initial condition:

$$\mathbf{C}(0) = \mathbf{C}_{\text{in}}.$$

Why MOL is Suitable for LLE Microchannel Problems

- LLE systems have strong transverse diffusion but dominant axial advection ($Pe \gg 1$), making MOL naturally aligned with flow physics.

- The method ensures numerical stability even in high-Péclet regimes by avoiding discretization in the advection direction.
- Interfacial gradients required for mass-transfer evaluations are computed accurately due to fine transverse resolution.

MODELLING OF THE SETUP

DERIVATION OF VELOCITY PROFILE FOR 1D STRATIFIED FLOW

1. Physical Setup

We consider a microchannel modeled as flow between two infinite parallel plates separated by a distance H .

- **Coordinate System:** Flow in x -direction, transverse variation in y -direction ($0 \leq y \leq H$), infinite in z -direction.
- **Phase 1 (Carrier):** Viscosity μ_1 , occupies $0 \leq y \leq h$.
- **Phase 2 (Solvent):** Viscosity μ_2 , occupies $h \leq y \leq H$.
- **Interface:** Located at $y = h$ (Dimensionless height $\alpha_s = h/H$).

2. Governing Equations

We start with the Navier-Stokes equation for an incompressible Newtonian fluid:

$$\rho \left(\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} \right) = -\nabla P + \mu \nabla^2 \mathbf{v} + \rho \mathbf{g} \quad (1)$$

Assumptions:

- 1) **Steady State:** $\partial/\partial t = 0$.
- 2) **Laminar Flow:** Inertial terms $\mathbf{v} \cdot \nabla \mathbf{v} \approx 0$.
- 3) **Unidirectional Fully Developed Flow:** $\mathbf{v} = [v_x(y), 0, 0]$. Thus, $\partial v_x/\partial x = 0$.
- 4) **Infinite Parallel Plates:** No variation in z ($\partial/\partial z = 0$).
- 5) **Constant Pressure Gradient:** $\partial P/\partial x = P_x$ (constant).

Applying these assumptions, the momentum equation simplifies to a balance between pressure and viscous forces:

$$\frac{d}{dy} \left(\mu \frac{dv_x}{dy} \right) = \frac{dP}{dx} \quad (2)$$

Since μ is constant within each phase:

$$\frac{d^2 v_x}{dy^2} = \frac{1}{\mu} P_x \quad (3)$$

3. General Solution

Integrating twice with respect to y yields the parabolic velocity profiles:

$$\text{Phase 1: } v_1(y) = \frac{P_x}{2\mu_1} y^2 + Ay + B \quad (0 \leq y \leq h) \quad (4)$$

$$\text{Phase 2: } v_2(y) = \frac{P_x}{2\mu_2} y^2 + Cy + D \quad (h \leq y \leq H) \quad (5)$$

4. Boundary Conditions

We determine the four constants (A, B, C, D) using physical boundary conditions:

1. No-Slip at Bottom Wall ($y = 0$):

$$v_1(0) = 0 \implies B = 0 \quad (6)$$

2. No-Slip at Top Wall ($y = H$):

$$v_2(H) = 0 \implies \frac{P_x}{2\mu_2} H^2 + CH + D = 0 \quad (7)$$

Rearranging for D : $D = -\frac{P_x H^2}{2\mu_2} - CH$.

3. Shear Stress Continuity at Interface ($y = h$):

$$\mu_1 \frac{dv_1}{dy} \Big|_h = \mu_2 \frac{dv_2}{dy} \Big|_h \quad (8)$$

Substituting derivatives:

$$\mu_1 \left(\frac{P_x}{\mu_1} h + A \right) = \mu_2 \left(\frac{P_x}{\mu_2} h + C \right) \quad (9)$$

Simplifying: $P_x h + \mu_1 A = P_x h + \mu_2 C \implies C = \frac{\mu_1}{\mu_2} A$.

4. Velocity Continuity at Interface ($y = h$):

$$v_1(h) = v_2(h) \quad (10)$$

5. Non-Dimensionalization

We introduce the dimensionless variables defined in the paper (Eq. 34-35):

$$Y = \frac{y}{H}, \quad \alpha_s = \frac{h}{H} \quad (11)$$

$$G_1 = \frac{P_x H^3}{\mu_1 \tilde{Q}_1}, \quad G_2 = \frac{P_x H^3}{\mu_2 \tilde{Q}_2} \quad (12)$$

where $\tilde{Q}_i$ is the flow rate per unit depth, and characteristic velocity is $v_{ch,i} = \tilde{Q}_i/H$.

The dimensionless velocities are $v_{1,dim} = v_1/v_{ch,1}$.

6. Final Dimensionless Equations

Substituting the solved constants and scaling variables yields the equations matching Eq. 36 and 37 in the paper:

Combined Velocity Profile

Phase 1 ($0 \leq y \leq \alpha_s$): The velocity profile v_1 is parabolic, strictly satisfying the no-slip condition at $y = 0$.

$$v_1(y) = \frac{G_1 y^2}{2} + \mu_2 \left[\frac{G_2 \tilde{Q}_r (\alpha_s^2 - 1) - G_1 \alpha_s^2}{2(\mu_1 - \alpha_s \mu_1 + \alpha_s \mu_2)} \right] y$$

Phase 2 ($\alpha_s \leq y \leq 1$): The velocity profile v_2 is parabolic, satisfying the no-slip condition at $y = 1$ and matching the velocity and shear stress of Phase 1 at the interface $y = \alpha_s$.

$$v_2(y) = \frac{G_2 y^2}{2} + \mu_1 \left[\frac{G_1 \alpha_s^2 - G_2 (\alpha_s^2 - 1) \tilde{Q}_r}{2 \tilde{Q}_r [(\alpha_s - 1) \mu_1 - \alpha_s \mu_2]} \right] y + \left[\frac{G_2 \alpha_s \tilde{Q}_r [(\alpha_s - 1) \mu_1 + \mu_2] - G_1 \alpha_s^2 \mu_1}{2 \tilde{Q}_r [(\alpha_s - 1) \mu_1 - \alpha_s \mu_2]} \right]$$

DERIVATION OF 1D STRATIFIED FLOW CONCENTRATION PDE

1. Species Mass Balance on a Control Volume

Consider a differential control volume of length Δx , height Δy , and unit width $W = 1$ within phase i . The general species balance equation is:

$$\text{Accumulation} = (\text{Rate In} - \text{Rate Out}) + \text{Generation} \quad (13)$$

Applying the assumptions for microfluidic extraction:

- **Steady State:** $\partial c / \partial t = 0$ (Accumulation = 0).
- **No Reaction:** Generation = 0.
- **Unidirectional Flow:** Velocity $\mathbf{v} = [v(y), 0, 0]$.
- **Negligible Axial Diffusion:** $Pe \gg 1$, so diffusion in the x -direction is negligible compared to advection.

The balance between advective flux in x and diffusive flux in y is:

$$J_{x,\text{adv}} \cdot \Delta y + J_{y,\text{diff}} \cdot \Delta x = 0 \quad (14)$$

Dividing by the volume $\Delta x \Delta y$ and taking the limit as $\Delta x, \Delta y \rightarrow 0$:

$$-\frac{\partial J_{x,\text{adv}}}{\partial x} - \frac{\partial J_{y,\text{diff}}}{\partial y} = 0 \quad (15)$$

Substituting the flux definitions $J_{x,\text{adv}} = v_i(y)c_i$ and $J_{y,\text{diff}} = -D_i \frac{\partial c_i}{\partial y}$:

$$-\frac{\partial}{\partial x}(v_i(y)c_i) - \frac{\partial}{\partial y}\left(-D_i \frac{\partial c_i}{\partial y}\right) = 0 \quad (16)$$

Since $v_i(y)$ is independent of x and D_i is constant:

$$v_i(y) \frac{\partial c_i}{\partial x} = D_i \frac{\partial^2 c_i}{\partial y^2} \quad (17)$$

2. Non-Dimensionalization

We introduce the dimensionless variables defined in Eq. 34 and 38 of the research paper:

$$X = \frac{x}{L} \quad Y = \frac{y}{H} \quad (18)$$

$$C_i = \frac{c_i}{c_{\text{ref}}} \quad v_i(y) = V_i(Y) \cdot v_{\text{ch},i} \quad (19)$$

The characteristic velocity scale for 1D flow (Eq. 34 of paper) is based on the flow rate per unit depth $\tilde{Q}_i$:

$$v_{\text{ch},i} = \frac{\tilde{Q}_i}{H} \quad (20)$$

Substituting these into Eq. 17:

$$\left[V_i \frac{\tilde{Q}_i}{H} \right] \frac{\partial(C_i c_{\text{ref}})}{\partial(XL)} = D_i \frac{\partial^2(C_i c_{\text{ref}})}{\partial(YH)^2} \quad (21)$$

Canceling c_{ref} and rearranging terms:

$$V_i \left(\frac{\tilde{Q}_i}{HL} \right) \frac{\partial C_i}{\partial X} = \frac{D_i}{H^2} \frac{\partial^2 C_i}{\partial Y^2} \quad (22)$$

Isolating the axial gradient $\frac{\partial C_i}{\partial X}$:

$$\frac{\partial C_i}{\partial X} = \frac{1}{V_i} \left(\frac{HL}{\tilde{Q}_i} \right) \left(\frac{D_i}{H^2} \right) \frac{\partial^2 C_i}{\partial Y^2} \quad (23)$$

Simplifying the geometric fractions:

$$\frac{\partial C_i}{\partial X} = \frac{1}{V_i} \left(\frac{L}{H} \right) \left(\frac{D_i}{\tilde{Q}_i} \right) \frac{\partial^2 C_i}{\partial Y^2} \quad (24)$$

Using the definition of the 1D Péclet number from Eq. 38 ($Pe_i^{1D} = \frac{\tilde{Q}_i}{D_i}$):

$$\frac{\partial C_i}{\partial X} = \frac{1}{V_i} \left(\frac{L}{H Pe_i^{1D}} \right) \frac{\partial^2 C_i}{\partial Y^2} \quad (25)$$

3. Final Governing Equations

This yields the final dimensionless forms presented in the paper (Eq. 39 and 40):

For Phase 1 (Carrier):

$$\frac{\partial c_1}{\partial x} = \left(\frac{1}{v_1} \right) \left(\frac{L}{H Pe_1^{1D}} \right) \frac{\partial^2 c_1}{\partial y^2}$$

For Phase 2 (Solvent):

$$\frac{\partial c_2}{\partial x} = \left(\frac{1}{v_2} \right) \left(\frac{L}{H Pe_2^{1D}} \right) \frac{\partial^2 c_2}{\partial y^2}$$

DERIVATION OF VELOCITY PROFILE FOR 2D STRATIFIED FLOW

1. Physical Setup and Governing Equations

Consider a rectangular microchannel of width H (y -direction), depth d (z -direction), and length L (x -direction). The aspect ratio is defined as $\lambda = H/d$.

- **Phase 1 (Carrier):** Occupies $0 \leq y \leq h$ (where $h = \alpha_s H$).
- **Phase 2 (Solvent):** Occupies $h \leq y \leq H$.
- The interface is a flat plane located at $y = \alpha_s H$.

We start with the Navier-Stokes equation for an incompressible Newtonian fluid:

$$\rho \left(\frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla \mathbf{v} \right) = -\nabla P + \mu \nabla^2 \mathbf{v} + \rho \mathbf{g} \quad (26)$$

Assumptions:

- 1) **Steady State:** $\frac{\partial}{\partial t} = 0$.
- 2) **Laminar Flow:** Low Reynolds number ($Re \ll 1$), inertial terms $\mathbf{v} \cdot \nabla \mathbf{v} \approx 0$.
- 3) **Fully Developed Flow:** $\mathbf{v} = [v_x(y, z), 0, 0]$. This implies $\frac{\partial v_x}{\partial x} = 0$.
- 4) **Constant Pressure Gradient:** $\frac{\partial P}{\partial x} = \text{constant}$.

Applying these assumptions, the momentum equation reduces to the Poisson equation:

$$\frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2} = \frac{1}{\mu} \frac{\partial P}{\partial x} \quad (27)$$

2. Non-Dimensionalization

We introduce the dimensionless variables (Eq. 19 in the paper):

$$Y = \frac{y}{H}, \quad Z = \frac{z}{d} \quad (28)$$

$$v = \frac{v_x}{v_{\text{ch}}}, \quad \lambda = \frac{H}{d} \quad (29)$$

Substituting these into the Laplacian operator:

$$\frac{1}{H^2} \frac{\partial^2 v}{\partial Y^2} + \frac{1}{d^2} \frac{\partial^2 v}{\partial Z^2} = \frac{1}{\mu v_{ch}} \frac{\partial P}{\partial x} \quad (30)$$

Multiplying the entire equation by H^2 :

$$\frac{\partial^2 v}{\partial Y^2} + \left(\frac{H}{d}\right)^2 \frac{\partial^2 v}{\partial Z^2} = \frac{H^2}{\mu v_{ch}} \frac{\partial P}{\partial x} \quad (31)$$

This yields the governing dimensionless equations (Eq. 21 and 22):

$$\text{Phase 1: } \frac{\partial^2 v_1}{\partial y^2} + \lambda^2 \frac{\partial^2 v_1}{\partial z^2} = G_1 \quad (32)$$

$$\text{Phase 2: } \frac{\partial^2 v_2}{\partial y^2} + \lambda^2 \frac{\partial^2 v_2}{\partial z^2} = G_2 \quad (33)$$

where G_i is the dimensionless pressure gradient parameter.

3. Boundary Conditions

The PDEs are subject to the following boundary conditions (Eq. 23-25):

A. No-Slip at Side Walls ($z = 0, z = 1$):

$$v_1(y, 0) = v_1(y, 1) = 0, \quad v_2(y, 0) = v_2(y, 1) = 0 \quad (34)$$

B. No-Slip at Top/Bottom Walls:

$$v_1(0, z) = 0, \quad v_2(1, z) = 0 \quad (35)$$

C. Interface Continuity ($y = \alpha_s$):

$$v_1(\alpha_s, z) = Q_r v_2(\alpha_s, z) \quad (\text{Velocity Continuity}) \quad (36)$$

$$\left. \frac{\partial v_1}{\partial y} \right|_{y=\alpha_s} = Q_1 \mu_r \left. \frac{\partial v_2}{\partial y} \right|_{y=\alpha_s} \quad (\text{Shear Stress Continuity}) \quad (37)$$

4. Analytical Solution: Partial Eigenfunction Expansion

We solve the PDE using a Fourier series expansion. Since the velocity vanishes at $z = 0$ and $z = 1$, we choose the basis functions $\sin(n\pi z)$. We assume a solution of the form:

$$v(y, z) = \sum_{n=1}^{\infty} \Phi_n(y) \sin(n\pi z) \quad (38)$$

Substituting this into the governing PDE transforms it into an Ordinary Differential Equation (ODE) for $\Phi_n(y)$:

$$\Phi_n''(y) - (n\pi\lambda)^2 \Phi_n(y) = g_n \quad (39)$$

where g_n are the Fourier coefficients of the constant pressure gradient G .

The general solution to this ODE involves hyperbolic functions:

$$\Phi_n(y) = C_1 \cosh(n\pi\lambda y) + C_2 \sinh(n\pi\lambda y) + \Phi_{\text{particular}} \quad (40)$$

5. Final Velocity Profiles

Applying the boundary conditions to determine the constants yields the final expressions (Eq. 26 and 27):

For Phase 1 ($0 \leq y \leq \alpha_s$):

$$v_1(y, z) = \sum_{n=1}^{\infty} \left[\frac{2G_1 ((-1)^n - 1)}{n^3 \pi^3 \lambda^2} \right] \times [A_{1n} \cosh(n\pi\lambda y) + A_{2n} \sinh(n\pi\lambda y) + 1] \sin(n\pi z)$$

For Phase 2 ($\alpha_s \leq y \leq 1$):

$$v_2(y, z) = \sum_{n=1}^{\infty} \left[\frac{2G_2 ((-1)^n - 1)}{n^3 \pi^3 \lambda^2} \right] \times [A_{3n} \cosh(n\pi\lambda y) + A_{4n} \sinh(n\pi\lambda y) + 1] \sin(n\pi z)$$

The constants $A_{1n}, A_{2n}, A_{3n}, A_{4n}$ are found by solving the 4×4 linear system resulting from the boundary conditions at $y = 0, \alpha_s, 1$.

DERIVATION OF 2D STRATIFIED FLOW CONCENTRATION PDE

1. Physical Setup

We consider a rectangular microchannel with the following dimensions:

- Length L (x-direction, flow direction)
- Width H (y-direction, transverse direction)
- Depth d (z-direction)

The flow is fully developed and laminar with velocity vector $\mathbf{v} = [v_x(y, z), 0, 0]$. The concentration $c(x, y, z)$ varies in all three dimensions.

2. Species Mass Balance on a 3D Control Volume

Consider a differential control volume (CV) of dimensions $\Delta x, \Delta y, \Delta z$ within one fluid phase. The general conservation law at steady state is:

$$\text{Rate In} - \text{Rate Out} = 0 \quad (41)$$

Flux Definitions: Let J_i be the molar flux in direction i .

- **x-direction (Advection):** $J_x = v_x c$ (since axial diffusion is negligible, $Pe \gg 1$).
- **y-direction (Diffusion):** $J_y = -D \frac{\partial c}{\partial y}$.
- **z-direction (Diffusion):** $J_z = -D \frac{\partial c}{\partial z}$.

Balance Equation: Summing the net rates across the faces (Flux $\times$ Area):

$$\underbrace{(J_x|_x - J_x|_{x+\Delta x}) \Delta y \Delta z}_{\text{Net x-flow}} + \underbrace{(J_y|_y - J_y|_{y+\Delta y}) \Delta x \Delta z}_{\text{Net y-flow}} + \underbrace{(J_z|_z - J_z|_{z+\Delta z}) \Delta x \Delta y}_{\text{Net z-flow}} = 0$$

Dividing the entire equation by the volume $V = \Delta x \Delta y \Delta z$:

$$\frac{J_x|_x - J_x|_{x+\Delta x}}{\Delta x} + \frac{J_y|_y - J_y|_{y+\Delta y}}{\Delta y} + \frac{J_z|_z - J_z|_{z+\Delta z}}{\Delta z} = 0 \quad (42)$$

Taking the limit as $\Delta x, \Delta y, \Delta z \rightarrow 0$, the terms become negative partial derivatives:

$$-\frac{\partial J_x}{\partial x} - \frac{\partial J_y}{\partial y} - \frac{\partial J_z}{\partial z} = 0 \quad (43)$$

Substituting the flux definitions:

$$-\frac{\partial(v_x c)}{\partial x} - \frac{\partial}{\partial y} \left(-D \frac{\partial c}{\partial y} \right) - \frac{\partial}{\partial z} \left(-D \frac{\partial c}{\partial z} \right) = 0 \quad (44)$$

Since v_x is independent of x and D is constant:

$$v_x \frac{\partial c}{\partial x} = D \left(\frac{\partial^2 c}{\partial y^2} + \frac{\partial^2 c}{\partial z^2} \right) \quad (45)$$

3. Non-Dimensionalization

We introduce the dimensionless variables defined in the research paper (Eq. 19):

$$X = \frac{x}{L} \quad Y = \frac{y}{H} \quad Z = \frac{z}{d} \quad (46)$$

$$V = \frac{v_x}{v_{ch}} \quad v_{ch} = \frac{Q}{H^2} \quad \lambda = \frac{H}{d} \quad (47)$$

Substituting these into Eq. 45:

- $\frac{\partial c}{\partial x} = \frac{1}{L} \frac{\partial c}{\partial X}$
- $\frac{\partial^2 c}{\partial y^2} = \frac{1}{H^2} \frac{\partial^2 c}{\partial Y^2}$
- $\frac{\partial^2 c}{\partial z^2} = \frac{1}{d^2} \frac{\partial^2 c}{\partial Z^2}$

The equation becomes:

$$\left(V \frac{Q}{H^2} \right) \frac{1}{L} \frac{\partial c}{\partial X} = D \left(\frac{1}{H^2} \frac{\partial^2 c}{\partial Y^2} + \frac{1}{d^2} \frac{\partial^2 c}{\partial Z^2} \right) \quad (48)$$

Rearranging to isolate $\frac{\partial c}{\partial X}$:

$$\frac{\partial c}{\partial X} = \frac{1}{V} \left(\frac{H^2 L}{Q} \right) D \left(\frac{1}{H^2} \frac{\partial^2 c}{\partial Y^2} + \frac{1}{d^2} \frac{\partial^2 c}{\partial Z^2} \right) \quad (49)$$

Distributing the constant term $\frac{DH^2L}{Q}$ into the bracket:

$$\frac{\partial c}{\partial X} = \frac{1}{V} \frac{DL}{Q} \left(\frac{\partial^2 c}{\partial Y^2} + \frac{H^2}{d^2} \frac{\partial^2 c}{\partial Z^2} \right) \quad (50)$$

Using the definition of Aspect Ratio $\lambda = H/d$:

$$\frac{\partial c}{\partial X} = \frac{1}{V} \frac{DL}{Q} \left(\frac{\partial^2 c}{\partial Y^2} + \lambda^2 \frac{\partial^2 c}{\partial Z^2} \right) \quad (51)$$

Using the definition of the 2D Péclet number (Eq. 28), $Pe^{2D} = \frac{Q}{DL}$:

$$\frac{DL}{Q} = \frac{1}{Pe^{2D}} \quad (52)$$

4. Final Governing Equations

This yields the final dimensionless PDEs (matching Eq. 29 and 30 in the paper):

For Phase 1 (Carrier, $0 \leq y \leq \alpha_s$):

$$\boxed{\frac{\partial c_1}{\partial x} = \left(\frac{1}{v_1} \right) \left(\frac{1}{Pe_1^{2D}} \right) \left[\frac{\partial^2 c_1}{\partial y^2} + \lambda^2 \frac{\partial^2 c_1}{\partial z^2} \right]}$$

For Phase 2 (Solvent, $\alpha_s \leq y \leq 1$):

$$\boxed{\frac{\partial c_2}{\partial x} = \left(\frac{1}{v_2} \right) \left(\frac{1}{Pe_2^{2D}} \right) \left[\frac{\partial^2 c_2}{\partial y^2} + \lambda^2 \frac{\partial^2 c_2}{\partial z^2} \right]}$$

DERIVATION OF VELOCITY PROFILE FOR CORE-ANNULAR FLOW

1. Physical Setup and Governing Equations

We consider a circular microchannel of radius R_0 and length L .

- **Core Fluid:** Viscosity μ_c , Flow rate Q_c , occupies $0 \leq r \leq R_i$.
- **Annular Fluid:** Viscosity μ_a , Flow rate Q_a , occupies $R_i \leq r \leq R_0$.
- **Interface:** Located at radius R_i (dimensionless $a = R_i/R_0$).

We start with the Navier-Stokes equation in cylindrical coordinates for the axial velocity $v_z(r, \theta, t)$:

$$\rho \left(\frac{\partial v_z}{\partial t} + \mathbf{v} \cdot \nabla v_z \right) = -\frac{\partial P}{\partial z} + \mu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_z}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 v_z}{\partial \theta^2} + \frac{\partial^2 v_z}{\partial z^2} \right]. \quad (53)$$

Assumptions:

- 1) **Steady State:** $\frac{\partial v_z}{\partial t} = 0$.
- 2) **Unidirectional Flow:** $\mathbf{v} = [0, 0, v_z]$.
- 3) **Fully Developed:** $\frac{\partial v_z}{\partial z} = 0$.
- 4) **Axisymmetric:** $\frac{\partial v_z}{\partial \theta} = 0$.

Applying these assumptions, the equation simplifies to a balance between viscous and pressure forces:

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_z}{\partial r} \right) = \frac{1}{\mu} \frac{\partial P}{\partial z} \quad (54)$$

2. Non-Dimensionalization

We introduce the dimensionless variables defined in the paper (Eq. 1):

$$R = \frac{r}{R_0} \quad (55)$$

$$v_c = \frac{v_{z,c}}{v_{c,ch}}, \quad v_a = \frac{v_{z,a}}{v_{a,ch}} \quad (56)$$

where the characteristic velocities are defined as superficial velocities: $v_{i,ch} = \frac{Q_i}{\pi R_0^2}$.

Substituting these into the governing ODE yields the dimensionless equations (Eq. 2 and 3):

$$\text{Core: } \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_c}{\partial r} \right) = G_c \quad (0 \leq r \leq a) \quad (57)$$

$$\text{Annulus: } \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_a}{\partial r} \right) = G_a \quad (a \leq r \leq 1) \quad (58)$$

Here, r denotes the dimensionless radius R , and G_i are the dimensionless pressure gradient parameters:

$$G_i = \frac{\pi R_0^4}{\mu_i Q_i} \left(\frac{\partial P}{\partial z} \right) \quad (59)$$

3. General Solution

Integrating the ODE $\frac{d}{dr}(r \frac{dv}{dr}) = Gr$ twice with respect to r :

$$v(r) = \frac{Gr^2}{4} + C_1 \ln(r) + C_2 \quad (60)$$

Applying this form to both phases:

$$v_c(r) = \frac{G_c r^2}{4} + B_1 \ln(r) + B_2 \quad (61)$$

$$v_a(r) = \frac{G_a r^2}{4} + A_1 \ln(r) + A_2 \quad (62)$$

4. Boundary Conditions

We determine the constants using the boundary conditions (Eq. 4-7):

1. Boundedness at Centerline ($r = 0$): The velocity must be finite at $r = 0$, so the logarithmic term in the core must vanish.

$$B_1 = 0 \implies v_c(r) = \frac{G_c r^2}{4} + B_2 \quad (63)$$

2. No-Slip at Wall ($r = 1$):

$$v_a(1) = 0 \implies \frac{G_a(1)^2}{4} + A_1 \ln(1) + A_2 = 0 \implies A_2 = -\frac{G_a}{4} \quad (64)$$

(Note: The shear stress continuity condition combined with the definitions of G_c and G_a implies $A_1 = 0$ for this specific formulation). Thus, the annular velocity becomes:

$$v_a(r) = \frac{G_a}{4}(r^2 - 1) \quad (65)$$

3. Velocity Continuity at Interface ($r = a$): The condition is $v_c(a) = Q_r v_a(a)$ (where $Q_r = Q_a/Q_c$). Substituting the expressions evaluated at $r = a$:

$$\frac{G_c a^2}{4} + B_2 = Q_r \left[\frac{G_a}{4}(a^2 - 1) \right] \quad (66)$$

Solving for the constant B_2 :

$$B_2 = \frac{Q_r G_a}{4}(a^2 - 1) - \frac{G_c a^2}{4} \quad (67)$$

5. Final Velocity Profiles

Substituting B_2 back into the core velocity equation gives the final analytical profiles matching Eq. 10 and 11 in the paper.

Annular Fluid ($a \leq r \leq 1$):

$$v_a(r) = \frac{G_a}{4}(r^2 - 1)$$

Core Fluid ($0 \leq r \leq a$):

$$v_c(r) = \frac{Q_r G_a}{4}(a^2 - 1) + \frac{G_c}{4}(r^2 - a^2)$$

DERIVATION OF CORE-ANNULAR FLOW CONCENTRATION PDE

1. Physical Setup and Geometry

We consider a circular microchannel with the following parameters:

- Radius R_0 and Length L .
- Cylindrical Coordinate System (r, z) .
- **Core Fluid:** Occupies $0 \leq r \leq R_i$.
- **Annular Fluid:** Occupies $R_i \leq r \leq R_0$.

The flow is fully developed ($v_z(r)$) and axisymmetric. Solute concentration $c(r, z)$ is assumed to be steady-state and axisymmetric.

2. Mass Balance on a Cylindrical Shell

Consider a differential control volume (CV) in the form of a cylindrical shell located at radius r with thickness Δr and length Δz .

Flux Definitions: Let J be the molar flux (moles per area per time).

- **Axial Flux (z):** Dominated by advection. $J_z = v_z(r)c$.
- **Radial Flux (r):** Dominated by diffusion. $J_r = -D \frac{\partial c}{\partial r}$.

Balance Equation (Rate In - Rate Out = 0): We sum the rates across the surfaces of the shell. Note that the area of the curved surface depends on r .

- Axial Area: $A_z = 2\pi r \Delta r$.
- Radial Area at r : $A_r = 2\pi r \Delta z$.

$$\underbrace{(J_z|_z - J_z|_{z+\Delta z}) 2\pi r \Delta r}_{\text{Net Axial Rate}} + \underbrace{J_r|_r (2\pi r \Delta z) - J_r|_{r+\Delta r} 2\pi (r + \Delta r) \Delta z}_{\text{Net Radial Rate}} = 0$$

Dividing by shell volume $V = 2\pi r \Delta r \Delta z$:

$$\frac{J_z|_z - J_z|_{z+\Delta z}}{\Delta z} + \frac{1}{r} \frac{(r J_r)|_r - (r J_r)|_{r+\Delta r}}{\Delta r} = 0 \quad (68)$$

Taking the limit as $\Delta z, \Delta r \rightarrow 0$ yields the divergence in cylindrical coordinates:

$$-\frac{\partial J_z}{\partial z} - \frac{1}{r} \frac{\partial}{\partial r} (r J_r) = 0 \quad (69)$$

Substituting the flux definitions:

$$-\frac{\partial (v_z c)}{\partial z} - \frac{1}{r} \frac{\partial}{\partial r} \left(r \left(-D \frac{\partial c}{\partial r} \right) \right) = 0 \quad (70)$$

Since v_z depends only on r and D is constant:

$$v_z(r) \frac{\partial c}{\partial z} = D \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial c}{\partial r} \right) \right] = D \left[\frac{\partial^2 c}{\partial r^2} + \frac{1}{r} \frac{\partial c}{\partial r} \right] \quad (71)$$

3. Non-Dimensionalization

We introduce the dimensionless variables defined in the research paper (Eq. 1 and 7):

$$Z = \frac{z}{L} \quad R = \frac{r}{R_0} \quad (72)$$

$$v = \frac{v_z}{v_{ch}} \quad v_{ch} = \frac{Q}{\pi R_0^2} \quad (73)$$

Substituting these into Eq. 71:

- $\frac{\partial c}{\partial z} = \frac{1}{L} \frac{\partial c}{\partial Z}$
- $\frac{\partial}{\partial r} = \frac{1}{R_0} \frac{\partial}{\partial R}$
- $\frac{\partial^2}{\partial r^2} = \frac{1}{R_0^2} \frac{\partial^2}{\partial R^2}$

$$(v \cdot v_{ch}) \frac{1}{L} \frac{\partial c}{\partial Z} = D \quad (74)$$

Simplifying and rearranging to isolate $\frac{\partial c}{\partial Z}$:

$$\frac{\partial c}{\partial Z} = \frac{1}{v} \left(\frac{L}{v_{ch}} \right) \frac{D}{R_0^2} \quad (75)$$

Grouping constants:

$$\text{Coeff} = \frac{DL}{v_{ch} R_0^2} = \frac{DL}{\left(\frac{Q}{\pi R_0^2} \right) R_0^2} = \frac{\pi DL}{Q} \quad (76)$$

Using the definition of the Péclet number (Eq. 14), $Pe = \frac{Q}{\pi DL}$:

$$\text{Coeff} = \frac{1}{Pe} \quad (77)$$

4. Final Governing Equations

This yields the final dimensionless forms matching Eq. 12 and 13 in the paper:

For Core Fluid ($0 \leq r \leq a$):

$$\frac{\partial c_c}{\partial z} = \left(\frac{1}{v_c} \right) \left(\frac{1}{Pe_c} \right) \left[\frac{\partial^2 c_c}{\partial r^2} + \frac{1}{r} \frac{\partial c_c}{\partial r} \right]$$

For Annular Fluid ($a \leq r \leq 1$):

$$\frac{\partial c_a}{\partial z} = \left(\frac{1}{v_a} \right) \left(\frac{1}{Pe_a} \right) \left[\frac{\partial^2 c_a}{\partial r^2} + \frac{1}{r} \frac{\partial c_a}{\partial r} \right]$$

EXTRACTION METRICS

Extraction Ratio

The Extraction Ratio (ER) measures the fraction of solute removed from the carrier phase by the solvent:

$$ER = \frac{Q_2 \bar{c}_{2,\text{out}}}{Q_1 c_{1,\text{in}}} \quad (78)$$

Here, $\bar{c}_{2,\text{out}}$ is the cross-sectional average concentration in the solvent at the outlet.

Extraction Efficiency

The Extraction Efficiency (E) compares the actual extraction to the maximum possible (equilibrium) extraction:

$$E = \frac{\bar{c}_{2,\text{out}} - \bar{c}_{2,\text{in}}}{c_{2,\text{eq}} - \bar{c}_{2,\text{in}}} \quad (79)$$

Here, $c_{2,\text{eq}}$ is the equilibrium concentration in the solvent based on the distribution coefficient.

RESULTS

A. 1-D Stratified Flow Velocity (Figure 4b from paper)

In the case of stratified flow, the less viscous fluid always contains the maximum axial velocity and flows faster than the more viscous fluid. This physical behavior is clearly visible in the velocity profiles shown in Fig. 1.

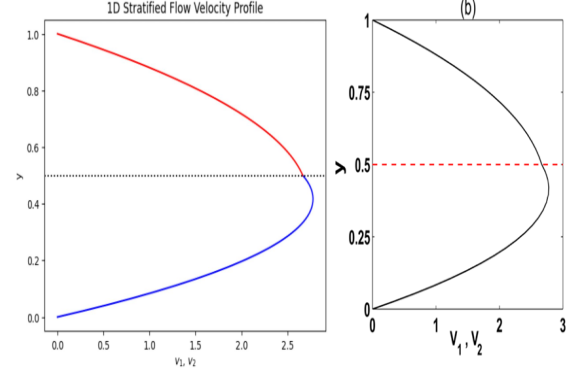


Fig. 1. Our Plot vs Paper's Plot

The two velocity profiles match extremely well in shape, curvature, and physical meaning: Peak velocity in the less viscous fluid, Lower velocity in the more viscous fluid, Smooth transition at the interface, Asymmetric parabolic shape, Correct no-slip conditions at the walls.

B. 2-D Stratified Flow Velocity (Plot by ourselves)

The plot is done by the velocity equations derived by simplifying Navier-Stoke's equation for 2-D stratified flow regime.

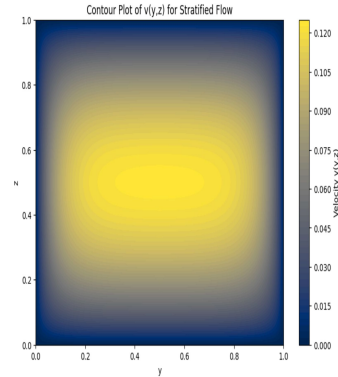


Fig. 2. Our plot

C. Core Annular Flow Velocity (Figure 4a from paper)

In the case of core-annular flow, the resistance to flow experienced by the core fluid is less than that experienced by the annular fluid, since the core is not in contact with the wall. Thus, the core fluid always contains the maximum of the velocity profile and flows with a higher average velocity than the annular fluid, irrespective of the viscosities of the

two fluids. This physical behavior is clearly visible in the velocity profiles shown in Fig. 3.

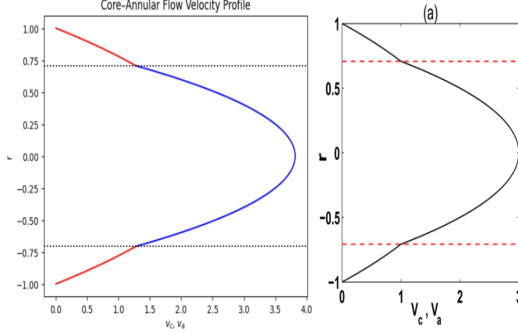


Fig. 3. Our Plot vs Paper's Plot

The velocity profiles obtained from our model show a strong qualitative match with the published results. Both plots exhibit the same asymmetric parabolic shape, a smooth interface transition, and correct boundary behavior—zero velocity at the walls and symmetry at the center. The core region consistently shows the highest velocity, as expected from theory. The only noticeable difference is the peak magnitude: the research paper reports a maximum velocity of about 3.0, whereas our plot reaches approximately 3.8. This small discrepancy likely arises from a difference in viscosity ratio or the pressure-gradient scaling, but it does not affect the overall consistency of the flow behavior.

D. Average concentration profile of liquid-liquid extraction in a microchannel with [a] stratified flow model, $k_s = 0.2703$, $\lambda = 0.1$ (Figure 7a from paper)

For fixed flow rates of both phases, the extraction performance was compared across different flow regimes. Figure 7(a) of research paper shows that the 1D stratified model (parallel plates) closely matches the 2D rectangular-channel model, with only a slight difference due to the added wall resistance in the 2D case, which leads to a higher pressure gradient compared to the 1D model. Despite this, the concentration profiles along the channel length agree well, indicating that for small aspect ratios ($\lambda = 0.1$), the simpler 1D model accurately captures the extraction behavior of the full 2D stratified system.

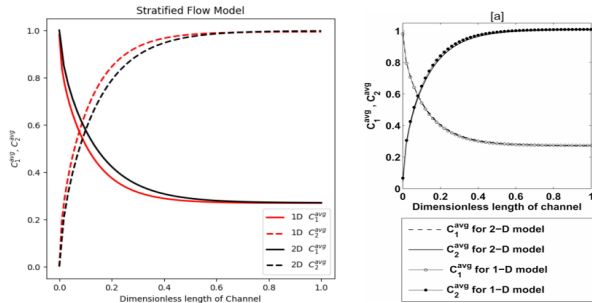


Fig. 4. Our Plot vs Paper's Plot

E. Average concentration profile of liquid-liquid extraction in a microchannel with [b] core annular flow model, $k_{ca} = 0.2703$, $\lambda = 0.1$ (Figure 7b from paper)

Figure 7(b) of the research paper shows the variation of the average concentration along the channel for core-annular flow. Two configurations are compared: the carrier fluid placed in the core and in the annulus. The flow rates of both phases are kept identical in the two cases, meaning the core flow rate in one setup becomes the annular flow rate in the other. The results clearly show that extraction is more effective when the carrier fluid occupies the annular region, due to its longer residence time.

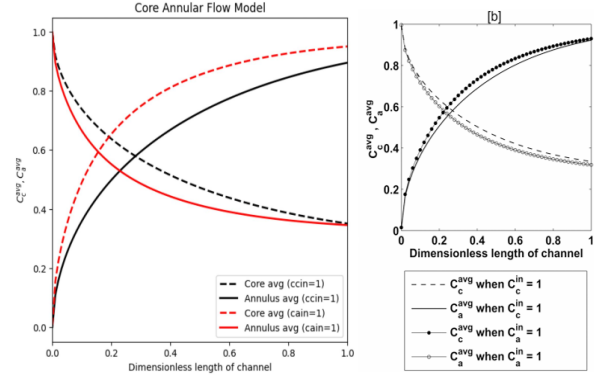


Fig. 5. Our Plot vs Paper's Plot

Comparing Figures 7(a) and 7(b) of the research paper, it is evident that stratified flow reaches equilibrium much faster than core-annular flow. As a result, the channel length required for equilibrium in core-annular flow is nearly four times larger than that needed for stratified flow under the same conditions.

F. Comparison of liquid liquid extraction performance in a microchannel based on extraction ratio (ER) for stratified and core annular flows [$k_s = k_{ca} = 0.2703, \lambda = 0.1$]. (Figure 8 from paper)

Figure 8 of the research paper compares the extraction ratio for stratified and core-annular flows. The simpler 1D stratified model predicts the extraction behavior accurately, which is expected given the small aspect ratio. Stratified flow shows the highest extraction performance under the chosen conditions. This is because the core holdup in both core-annular configurations is below the critical value ($\alpha_{core,crit} = 0.796$), giving stratified flow a larger interfacial area. Additionally, its characteristic time ratio is much smaller ($\tau = 0.345$) compared with core-annular flow ($\tau = 2.193$ for carrier-in-core and $\tau = 0.719$ for carrier-in-annulus). Consequently, stratified flow achieves greater extraction for the same channel length.

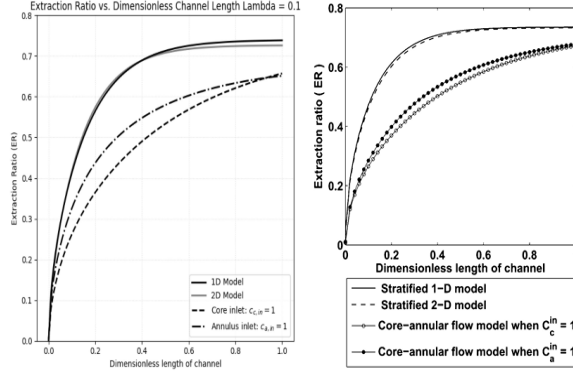


Fig. 6. Our Plot vs Paper's Plot

Both figures show the extraction ratio increasing along the dimensionless channel length for the four cases considered: 1D stratified flow, 2D stratified flow, core–annular with the carrier in the core, and core–annular with the carrier in the annulus. The trends in the reproduced plot closely match those in the reference figure. In both plots, the 1D and 2D stratified curves nearly overlap, confirming that the 1D model captures the extraction behavior well for small aspect ratios. Likewise, the two core–annular curves lie below the stratified curves, with the annulus–inlet case outperforming the core–inlet case, consistent with the results reported in the paper.

G. Liquid-liquid extraction performance based on extraction ratio (ER) along the dimensionless length of channel for different aspect ratios. (a) $\lambda = 1$ (b) $\lambda = 2.5$ [$k_s = k_{ca} = 0.2703$]. (Figure 9 from paper)

Figure 9 from the research paper and the recreated plot show very strong agreement across all four flow configurations and both aspect ratios. In both images, the extraction ratio increases rapidly near the inlet and gradually approaches a common asymptotic limit, demonstrating consistent mass-transfer behavior. The stratified 1-D and 2-D curves almost overlap in both figures, confirming that wall effects remain negligible for the aspect ratios studied. The ordering of the core–annular cases is also identical: the annulus-inlet configuration exhibits higher extraction compared to the core-inlet case due to its longer residence time and larger interfacial area. Furthermore, the influence of aspect ratio is captured similarly in both figures— $\lambda = 2.5$ leads to faster equilibration than $\lambda = 1$, reflecting reduced transverse diffusion length. The close correspondence in curve shapes, asymptotic ER values, and relative model behavior verifies that the recreated model faithfully reproduces the trends reported in the original Figure 9.

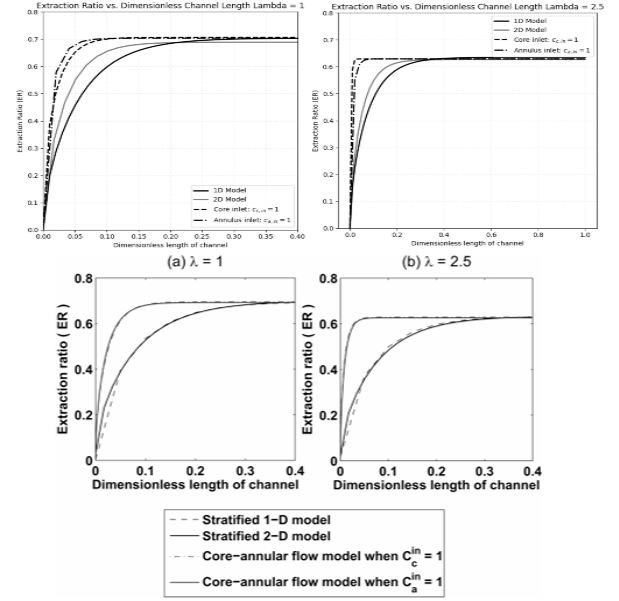


Fig. 7. Our plot (above) vs Paper's Plot(Below)

H. Liquid-liquid extraction performance at fixed pressure gradient and holdup for core–annular and stratified flow in a microchannel. [$\alpha_s = \alpha_{ca} = 0.5$, $\nabla P = -1185 \text{ Pa m}^{-1}$, $k_s = k_{ca} = 0.8772$]. (Figure 10 from paper)

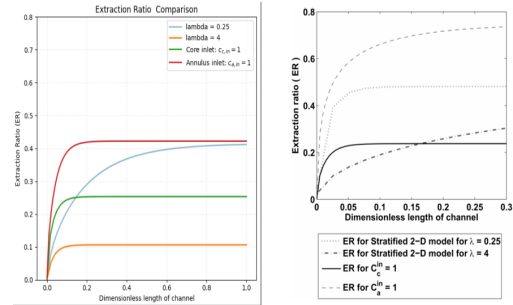


Fig. 8. Our plot (above) vs Paper's Plot(Below)

MASS BALANCE VALIDATION

I. MASS-BALANCE VERIFICATION

At each axial location x , the solute flows in the two phases and the cumulative interfacial transfer are used to compute the mass-balance residual

$$R(x) = \dot{M}_{\text{in}} - [Q_1 \bar{c}_1(x) + Q_2 \bar{c}_2(x)] - \int_0^x J_{\text{int}}(s) ds, \quad (80)$$

where $\dot{M}_{\text{in}} = Q_1 c_{1,\text{in}}$ is the inlet solute flow, $\bar{c}_i(x)$ is the cross-sectional average concentration of phase i , and $J_{\text{int}}(x)$ is the interfacial flux,

$$J_{\text{int}}(x) = -D_1 \left. \frac{\partial c_1}{\partial n} \right|_{\Gamma} = -D_2 \left. \frac{\partial c_2}{\partial n} \right|_{\Gamma}. \quad (81)$$

A consistent numerical solution yields $R(x) \approx 0$ along the channel. The cumulative solute-flow plots show phase 1 losing solute and phase 2 gaining solute, while the total remains nearly constant. The residual plot confirms conservation, with absolute errors on the order of 10^{-9} – 10^{-8} , well within the acceptable tolerance.

1-D stratified flow regime mass balance

Using equation 80, 81, 82, 83

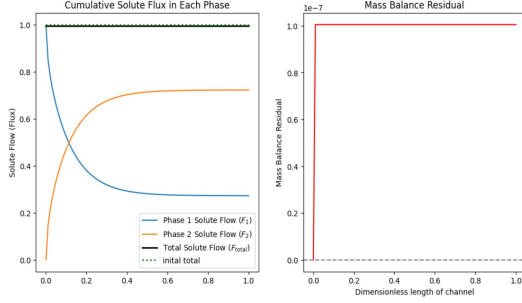


Fig. 9.

1-D Mass Balance Results

Inlet Total Solute Flow ($F_1(0) + F_2(0)$): 0.9745125424

Outlet Total Solute Flow ($F_1(L) + F_2(L)$): 0.9745128310

Total Flux Change ($F(0) - F(L)$): $-2.8860226431 \times 10^{-7}$

Maximum Absolute Residual ($R(x)$): $2.9898380516 \times 10^{-7}$

2-D stratified flow regime mass balance

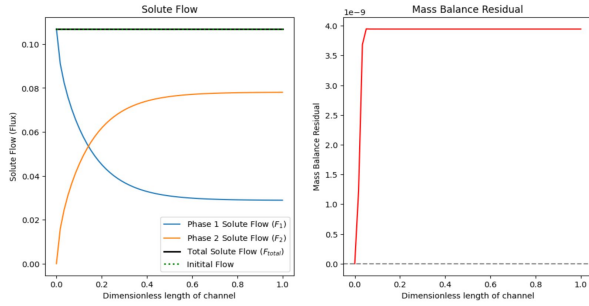


Fig. 10.

2-D Mass Balance Results Inlet Total Solute Flow ($F_1(0) + F_2(0)$): 0.1069268149

Outlet Total Solute Flow ($F_1(L) + F_2(L)$): 0.1069268189

Total Flux Change ($F(0) - F(L)$): $-3.9445400279 \times 10^{-9}$

Maximum Absolute Residual ($R(x)$): $3.9478028485 \times 10^{-9}$

Core Annular flow regime mass balance

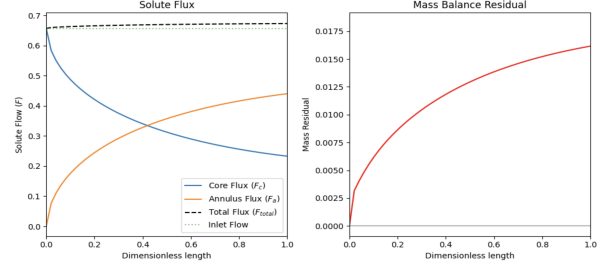


Fig. 11. $C_{in}^c = 1$

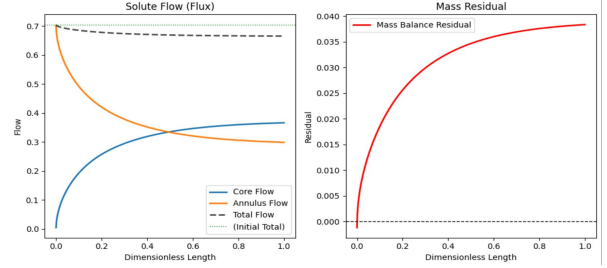


Fig. 12. $C_{in}^a = 1$

KEY LEARNINGS

- **Nondimensional groups are decisive:** Péclet number, viscosity ratio and holdup immediately identify dominating transport and flow regimes.
- **Model hierarchy:** 1D stratified models suffice for small aspect ratios ($\lambda \ll 1$); 2D/cylindrical models are needed when wall effects or curvature matter.
- **Interface controls transfer:** Accurate evaluation of $\partial c / \partial n$ at the interface is far more impactful on ER/E than bulk averages.
- **Analytical benchmarks speed validation:** Closed-form profiles (parabolic, series solutions) are invaluable for verifying numerical implementations.
- **Verification is critical:** Simple diagnostics (global mass balance, cumulative flux plots) catch subtle implementation errors efficiently.

ACKNOWLEDGEMENTS

The group gratefully acknowledge the guidance and support of Prof. Karthik Pushpavanam Subramaniam. His insightful suggestions on mass transfer and diffusion greatly improved the clarity of the study.

PLEASE ACCESS THE LINK TO THE CODE

Code Repository: Python Code and Plots